

Project Design Phase
Problem – Solution Fit Template

Date	02 August 2026
Team ID	
Project Name	Metro Ticket Booking in Servicenow
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Metro Ticket Booking with QR Code Generation project addresses the difficulties faced by metro commuters while purchasing tickets. Traditional ticket booking methods involve long queues, manual ticket issuance, and the risk of losing paper tickets. To overcome these challenges, this project provides a digital ticket booking system developed on the ServiceNow platform.

Users can select the source station, destination station, passenger type, and number of tickets through the Service Catalog. The system automatically calculates the fare, stores booking details, and generates a QR-code-based digital ticket for verification at metro stations. This improves convenience, reduces waiting time, and provides a faster, secure, and contactless ticketing experience.

Purpose:

- Solve the problem of long waiting queues and manual metro ticket booking.
- Provide a digital platform for booking metro tickets anytime through the ServiceNow Service Catalog.
- Automatically calculate ticket fares based on the selected stations and passenger type.
- Generate secure QR-code-based tickets for quick verification at metro entry gates.
- Reduce paper ticket usage and improve operational efficiency through digital ticket management.
- Enhance the commuter experience by offering a fast, reliable, and user-friendly booking process.

Template:

Metro Ticket Booking with QR Code Generation – Problem–Solution Fit Template

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS Who is your customer? Metro commuters including daily travellers, students, working professionals and tourists. Age group: 16+ People who use metro regularly for travel.	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices of solutions? • Long queues at metro ticket counters. • Time-consuming manual ticket booking. • Risk of losing paper tickets. • Need for exact cash and no digital payment options. • No direct QR-based entry for faster verification.	5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem or need to get the job done? • Manual ticket booking at counters. • Smart cards (limited to registered users). • Mobile apps of metro authorities (not always integrated or available). • Paper tickets.	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Which jobs-to-be-done (or problems) do you address for your customers? • Need to book metro tickets quickly and easily. • Need a secure and valid ticket for metro travel. • Need to avoid long waiting time at counters. • Need digital proof of booking for travel.	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job? • High footfall and limited ticket counters. • Manual processes increase waiting time. • Lack of integrated digital ticketing system. • Paper ticket handling and verification is slow. • No centralized system to manage bookings and history.	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? • Stand in queues and buy paper tickets. • Use metro cards if available. • Plan travel in advance to avoid rush. • Sometimes avoid peak hours. • Use third-party apps if available.	Focus on J&P, tap into BE, understand
Identify strong TR & EM	3. TRIGGERS TR What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news. • Need to travel by metro. • Seeing long queues at ticket counters. • News or ads about digital ticketing and QR tickets. • Recommendation from friends or family.	10. YOUR SOLUTION SL If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour. Our Solution: A ServiceNow-based Metro Ticket Booking System that allows users to: • Select source & destination stations, passenger type and number of tickets. • Automatically calculate fare. • Book tickets through Service Catalog. • Generate QR code for each booking. • Store booking history in the system. • Enable quick QR verification at metro entry gates.		Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? • Before: Frustrated, annoyed, time-consuming, worried about losing tickets. • After: Happy, satisfied, saves time, convenient, confident with digital ticket.	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE What kind of actions do customers take online? • Search metro ticket booking online. • Use metro authority website or mobile apps. • Social media, digital ads, recommendations. • Use ServiceNow Service Catalog portal. 8.2 OFFLINE What kind of actions do customers take offline? • Visit metro stations and ticket counters. • Ask metro staff for ticket information. • Use metro cards at stations.		